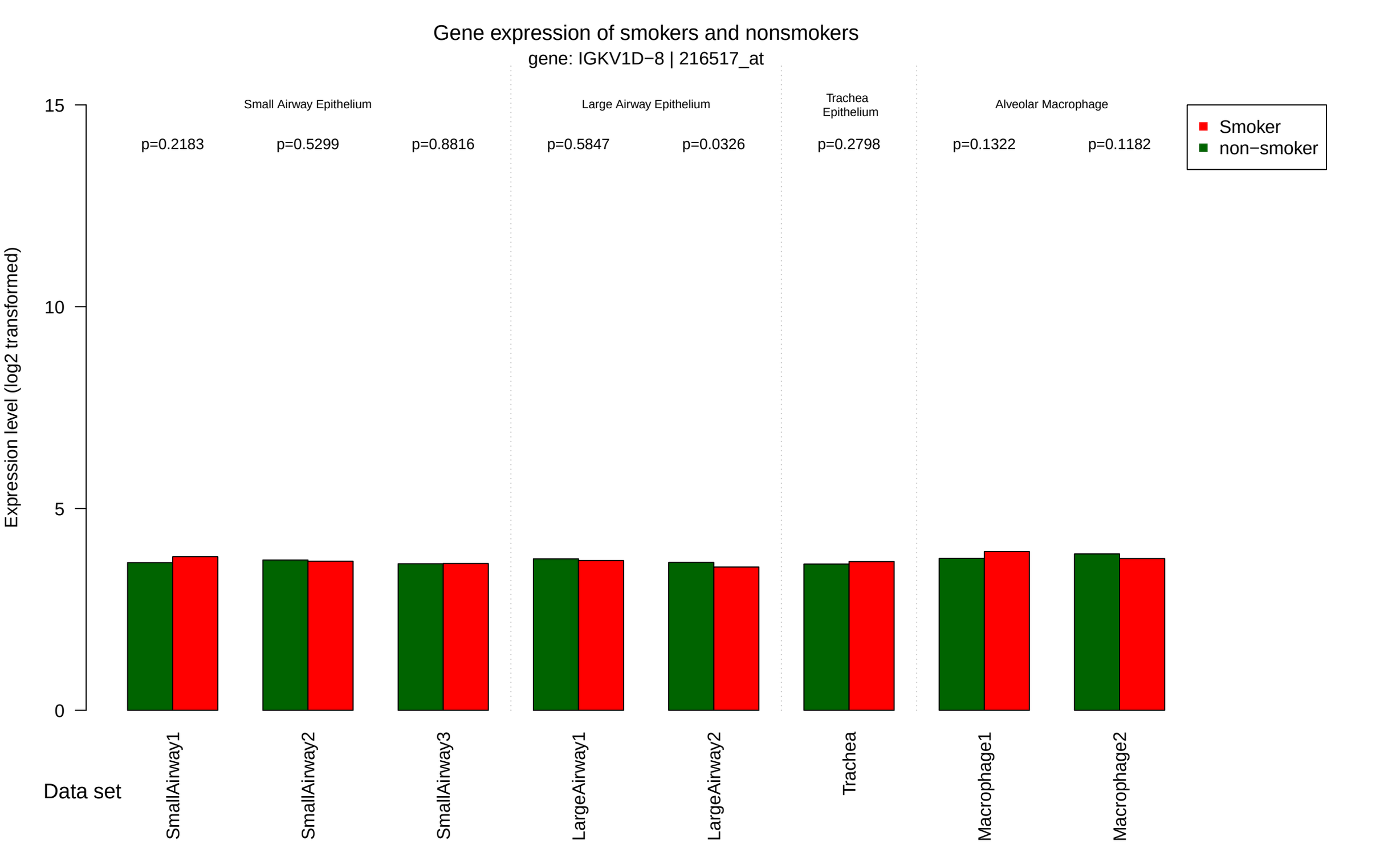
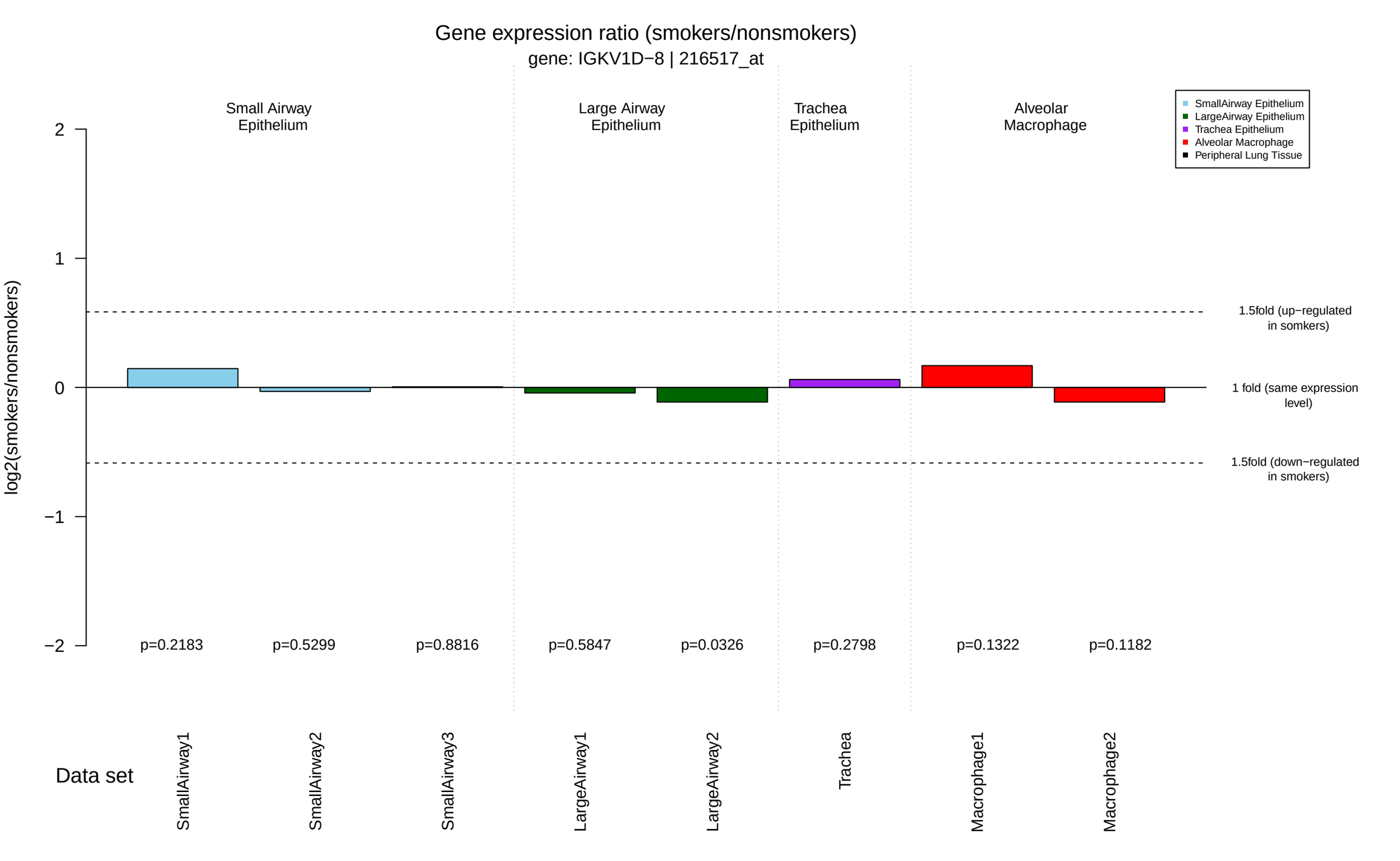
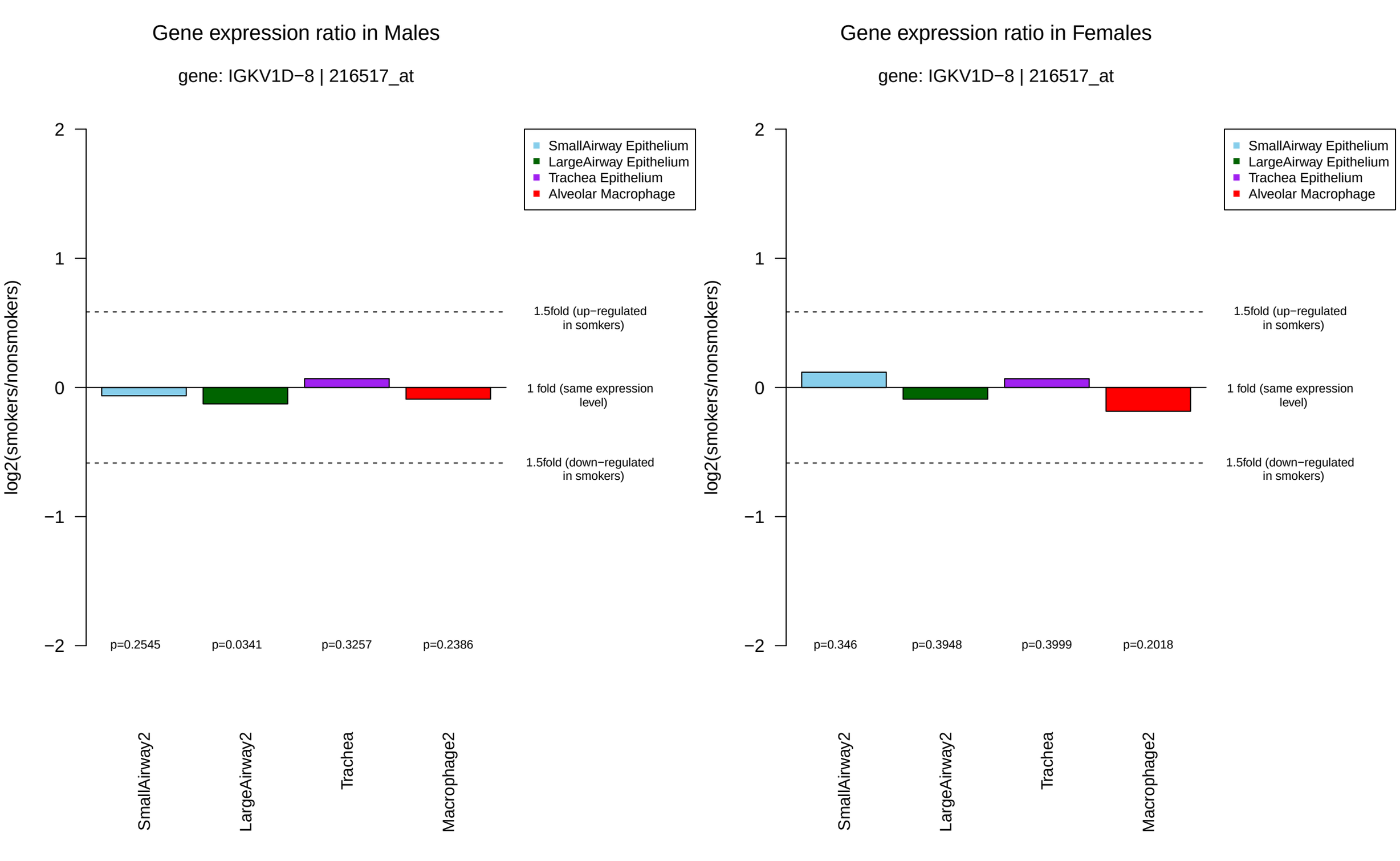
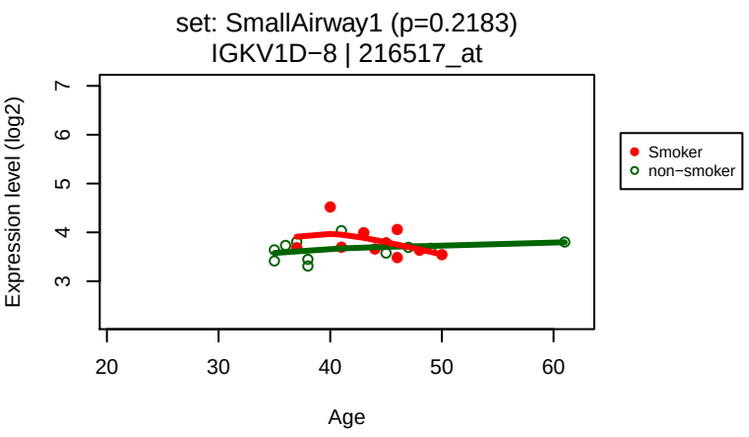
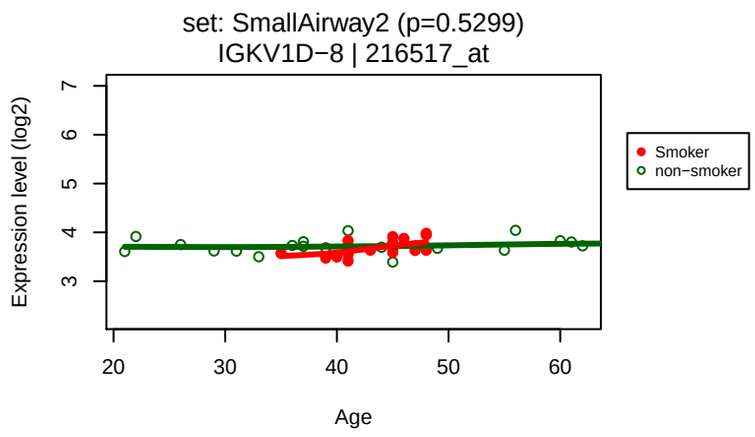
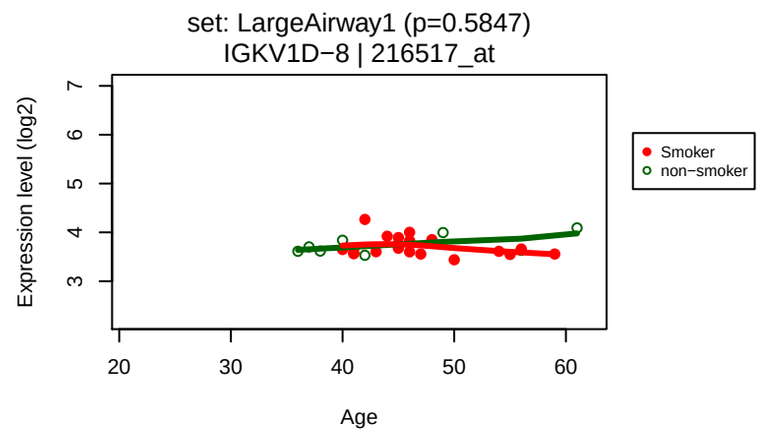

Gene expression of smokers and nonsmokers
gene: IGKV1D−8 | 216517_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2183
p=0.5299
p=0.8816
p=0.5847
p=0.0326
p=0.2798
p=0.1322
p=0.1182
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

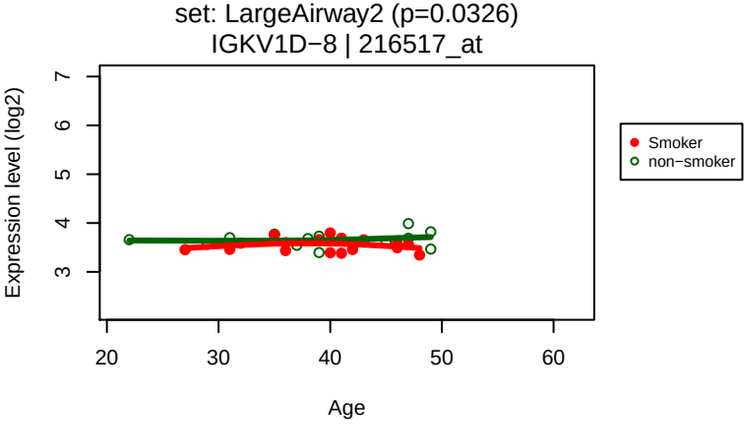

Gene expression ratio (smokers/nonsmokers)
gene: IGKV1D−8 | 216517_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2183
p=0.5299
p=0.8816
p=0.5847
p=0.0326
p=0.2798
p=0.1322
p=0.1182
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

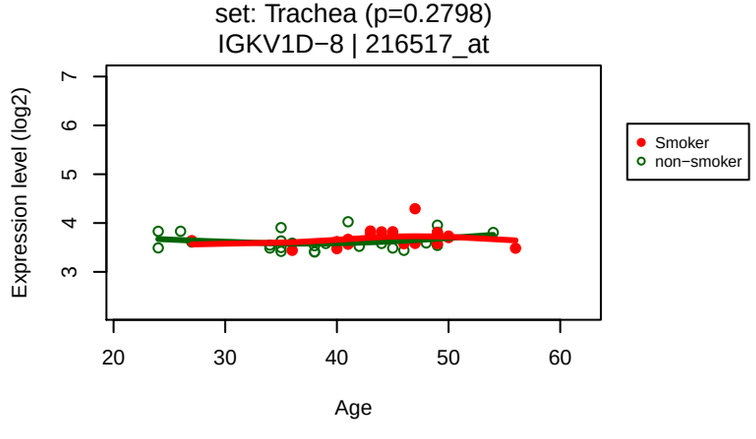

Gene expression ratio in Males
gene: IGKV1D−8 | 216517_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2545
p=0.0341
p=0.3257
p=0.2386
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IGKV1D−8 | 216517_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.346
p=0.3948
p=0.3999
p=0.2018
SmallAirway2
LargeAirway2
Trachea
Macrophage2

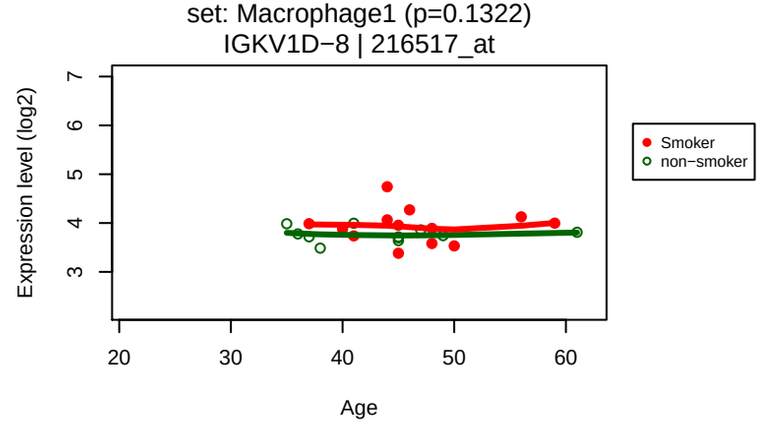

set: SmallAirway1 (p=0.2183)
IGKV1D−8 | 216517_at
Expression level (log2)
Age
Smoker
non−smoker

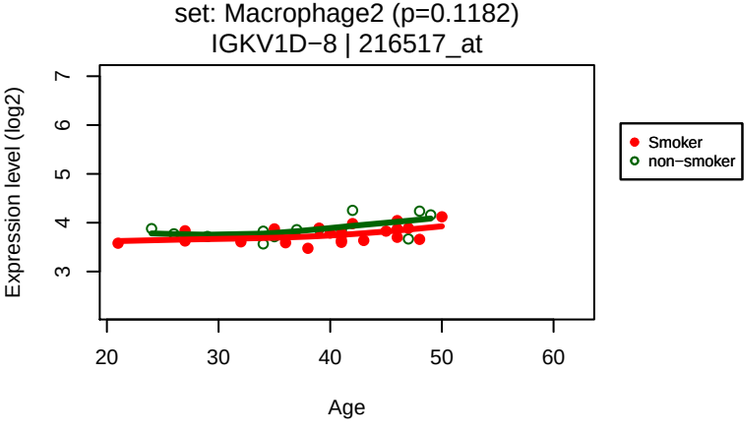

set: SmallAirway2 (p=0.5299)
IGKV1D−8 | 216517_at
Expression level (log2)
Age
Smoker
non−smoker

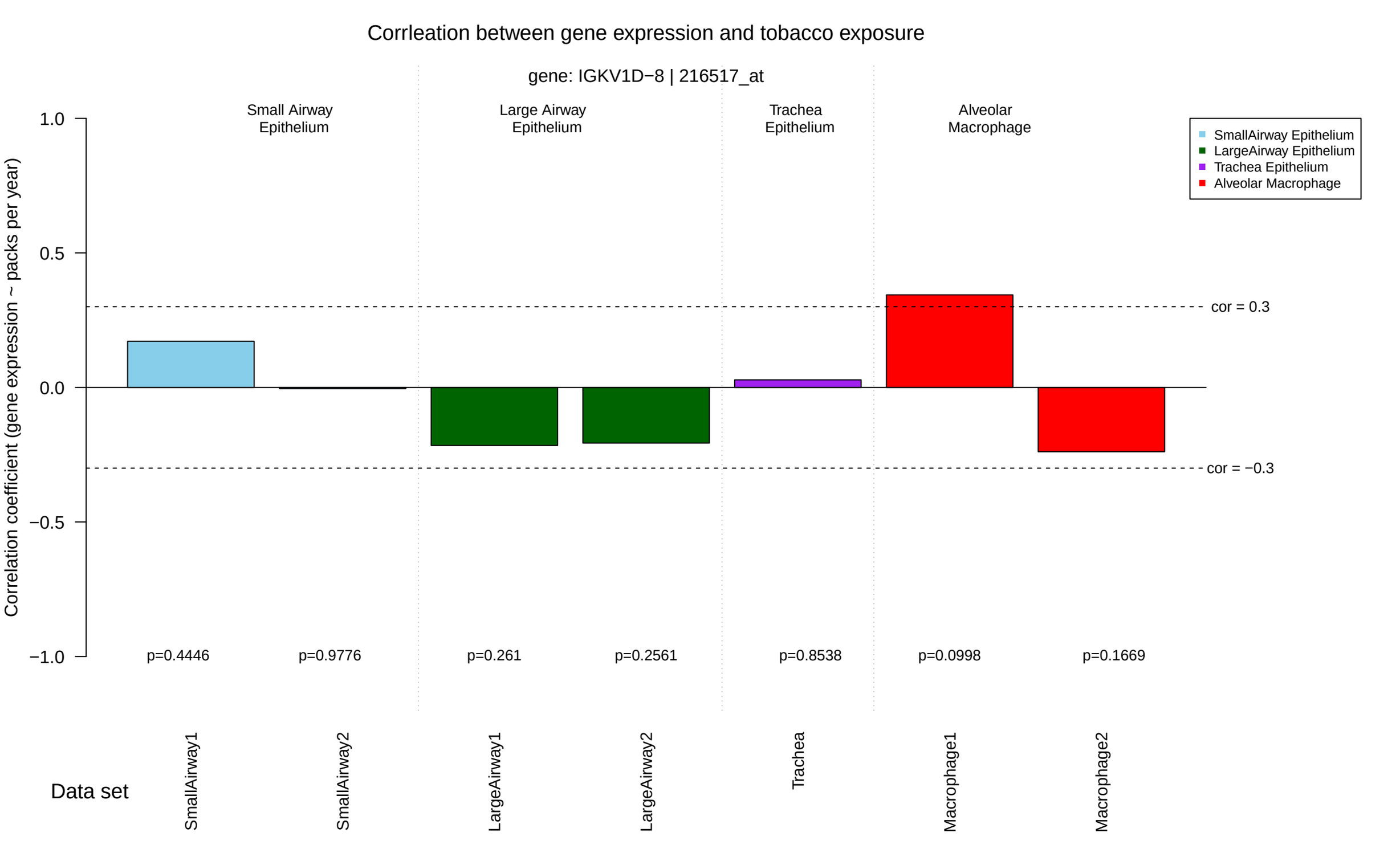

set: LargeAirway1 (p=0.5847)
IGKV1D−8 | 216517_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0326)
IGKV1D−8 | 216517_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2798)
IGKV1D−8 | 216517_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1322)
IGKV1D−8 | 216517_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1182)
IGKV1D−8 | 216517_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IGKV1D−8 | 216517_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.4446
p=0.9776
p=0.261
p=0.2561
p=0.8538
p=0.0998
p=0.1669
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2